

Design and Implementation of a Database Management System for Courier Parcel Booking and Tracking Management System

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DBMS

Introduction

Courier companies process high parcel volumes every day. Manual records create errors, delays, and inconsistent updates across branches. A centralized DBMS connects bookings, tracking, routes, deliveries, charges, and payments.

PARCEL LIFECYCLE

Booking → Tracking → Delivery → Payment



Problem Statement

- Inaccurate parcel tracking
- Duplicate or missing records
- Charge and payment errors
- No central movement history
- Delays are hard to identify
- Reporting is time-consuming
- Inconsistent branch / hub data

PROPOSED SOLUTION

**One relational database.
One reliable parcel record.**

Centralize customers, parcels, bookings, tracking events, charges, and payments—while enforcing consistent rules across every branch and hub.

Objectives

01

Centralize courier operations

02

Maintain customer & parcel data

03

Issue unique tracking numbers

04

Track movement end-to-end

05

Manage hubs, routes & resources

06

Record delivery attempts

07

Calculate courier charges

08

Maintain payment records

09

Generate management reports

Scope of the System

Customer Management	Registration • booking history
Parcel Management	Booking • tracking number • weight / dimensions • status
Courier Operations	Branches / hubs • routes • vehicles • delivery agents
Tracking & Delivery	Scan events • route movement • updates • failed attempts
Financial Management	Charge calculation • payments • revenue reports

Users & Functional Requirements

Customer Book • track • view payment	Branch Staff Book • scan • update movement	Delivery Agent Deliver • record attempts	Admin / Manager Manage network • reporting	Database Administrator Security • maintenance
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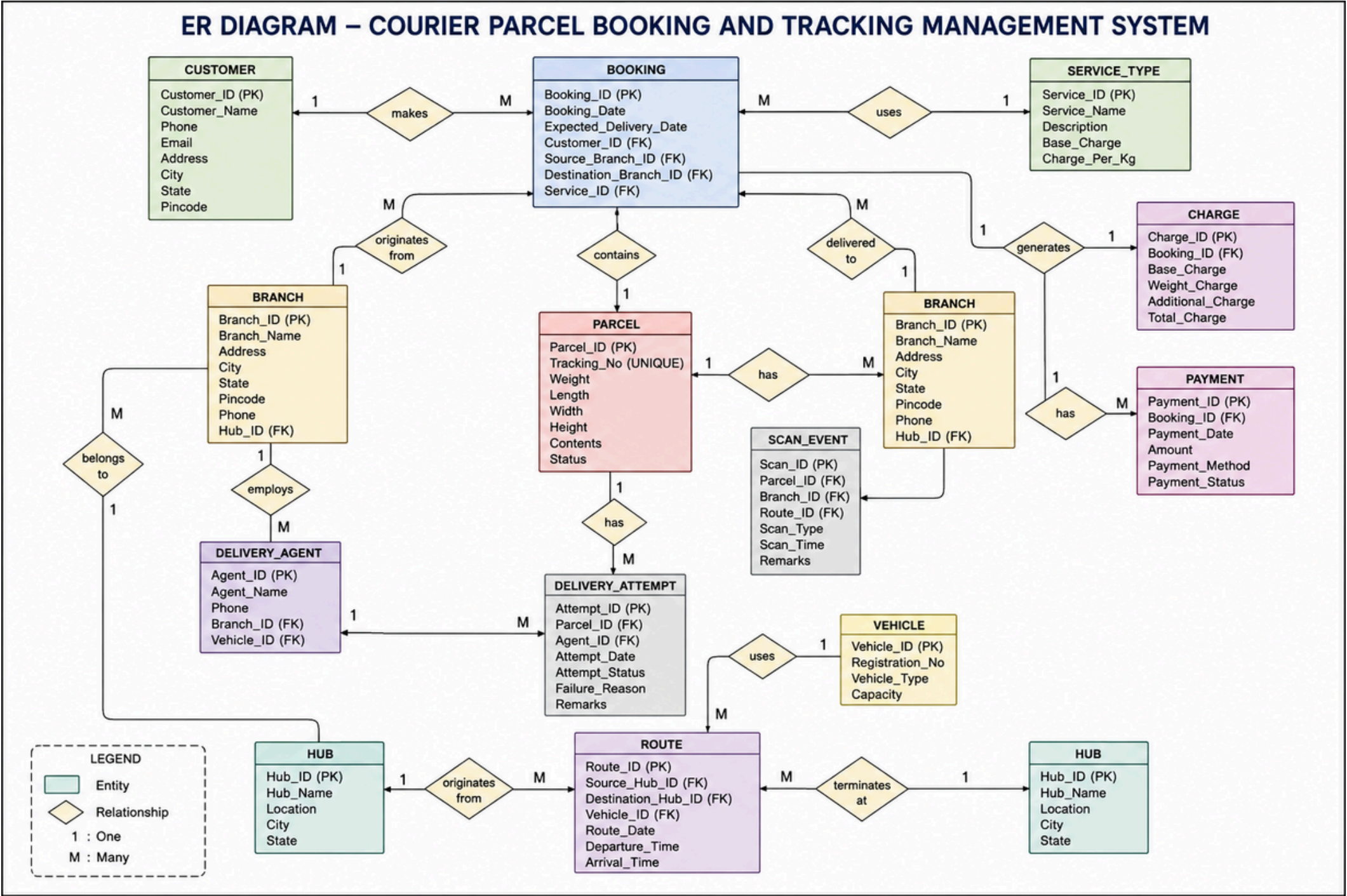
FUNCTIONAL CORE

Customer registration • Parcel booking • Charge calculation • Tracking and scanning • Route movement • Agent assignment • Delivery updates • Failed-delivery handling • Payment management • Report generation

Major Entities

● Customer	● Branch	● Hub	● Service Type
● Parcel	● Booking	● Route	● Scan Event
● Vehicle	● Delivery Agent	● Delivery Attempt	● Charge
● Payment			

Entity-Relationship Diagram



Entity-Relationship Diagram of Courier Parcel Booking and Tracking System

Initial Relational Schema

CUSTOMER	BRANCH	HUB	SERVICE_TYPE	PARCEL
Customer_ID PK Name • Phone • Email Address • City • State • Pincode	Branch_ID PK Name • Address • City • State Pincode • Phone • Hub_ID FK	Hub_ID PK Name • Location City • State	Service_ID PK Name • Description Base_Charge • Charge_Per_Kg	Parcel_ID PK Tracking_No UNIQUE Weight • Dimensions • Contents • Status

OTHER TABLES

Booking • Route • Scan_Event • Vehicle • Delivery_Agent • Delivery_Attempt • Charge • Payment

PK = Primary Key FK = Foreign Key

Business Rules & Future Work

BUSINESS RULES

- Unique tracking number per parcel
- Parcel weight must be greater than zero
- Charges positive; payments cannot exceed charges
- Delivered parcels accept no further scans
- Scan events follow a valid status sequence
- Attempts reference a valid parcel and agent
- All foreign keys preserve referential integrity

FUTURE WORK

- Normalize from 1NF to 3NF
- Complete data dictionary, DDL, and constraints
- Load sample data; build SQL queries and views
- Generate operational and management reports
- Develop Python / Java application connectivity
- Implement booking, pricing, tracking, scanning, assignment, updates, and testing

Thank You

Questions & Suggestions?

Q&A